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Quantum Support Vector Machine for Fraud Detection

1st Liang Ren
China Academy of Electronics and Information Technology
Beijing, China
renliang23@cetc.com.cn

2nd Xuesong Zhang*
China Academy of Electronics and Information Technology
Beijing, China
zhangxuesong@cetc.com.cn

***Abstract*—Quantum algorithms have demonstrated extraordinary potential in numerous fields. The demand for fraud detection in credit cards is growing increasingly, which calls for more efficient algorithms. This work adopts the detection method of Quantum Support Vector Machine (QSVM) to improve the accuracy of credit card fraud detection. Meanwhile, Quantum-Synthetic Minority Oversampling Technique (Quantum-SMOTE) is utilized to address the issue of imbalanced credit card datasets. The QSVM model maps classical data to a high-dimensional Hilbert space through quantum feature mapping, thereby enhancing the separability of features. Then, quantum kernel computation is performed to calculate the inner products between features for model training. The research results show that the QSVM achieves the best overall performance, which indicates that QSVM has significant potential in handling credit card fraud detection.**

Index Terms-quantum machine learning; quantum support vector machine; Quantum-SMOTE; credit card fraud detection

I. Introduction

In recent years, the rapid development of information technologies such as the Internet has seen financial technology, represented by artificial intelligence, increasingly penetrate the economic and social development sectors. The digital lifestyle driven by fintech not only influences consumer behavior but also transforms traditional financial models. With the continuous expansion of global trade and the exponential growth of internet technology, credit cards have become increasingly vital as a financial payment method. According to the "China Bank Card Industry Development Blue Book (2023)" released by the China Banking Association, by the end of 2022, China had issued a total of 798 million credit cards, with overdue credit balances exceeding six months reaching 86.58 billion yuan – marking a 0.6% year-on-year increase [1].

Manual verification proves impractical for credit card fraud detection, necessitating fully automated machine learning-based systems [2]. Since the 1980s, with the rapid development and widespread application of machine learning technology, an increasing number of researchers have begun applying machine learning techniques to credit card fraud detection, achieving abundant research outcomes. Jiang Hongxun et al. systematically reviewed the research progress of machine learning in credit card fraud detection since 2000, covering key aspects such as feature engineering, model algorithms, and evaluation metrics [3]. Xu Yunhao proposed a solution based on generative adversarial networks (GAN) to alleviate category imbalance in credit card transaction data, while optimizing features of the original dataset through voting-based feature selection [4]. Yang Fan et al. achieved high-precision detection of fraudulent transactions by jointly optimizing topological patterns and temporal transaction patterns in fraud prediction networks [5]. Xie et al. proposed an improved LSTM network architecture that effectively captures temporal dynamic features of user behavior in credit card transactions, thereby extracting more refined and comprehensive behavioral patterns. Experimental results based on real transaction data from a financial institution and public datasets demonstrated the significant effectiveness of this method [6]. Awoyemi et al. systematically evaluated the performance of four machine learning algorithms (RBF neural network, decision tree, naive Bayes, and SVM) in credit card fraud detection. By constructing classification models based on real credit card transaction data, they focused on comparing differences among various algorithms in terms of classification accuracy, sensitivity, and adaptability to imbalanced datasets [7].

However, applying machine learning algorithms to tackle credit card fraud presents multiple challenges [8]. With fraudulent transactions constituting only a small proportion of cases, the data becomes severely imbalanced. This imbalance causes models to predominantly predict the majority class during predictions. For imbalanced classification tasks, accurately identifying minority classes proves critical. Furthermore, in the era of financial big data, how to reduce training sample sizes while maintaining detection accuracy — thereby decreasing model complexity and computational costs—has become a key research focus [9].

The computational power of electronic computers is approaching its physical limits, while artificial intelligence and other fields are demanding higher processing capabilities. To address this challenge, researchers are advancing both the scale of computing systems to enhance parallel processing and exploring new computing devices like quantum computers. By leveraging quantum entanglement and superposition phenomena, quantum computers can dramatically improve parallel processing efficiency, demonstrating immense computational potential [10].

To address these challenges, this work proposes a quantum machine learning solution: First, Quantum-SMOTE technology is employed to resolve data imbalance issues. By conducting quantum swap tests and quantum rotation, high-quality synthetic minority class samples are generated, effectively improving data distribution. Then, the QSVM model is applied for training, leveraging the

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advantages of quantum kernel methods to significantly reduce computational complexity while maintaining classification performance. This combined approach not only enhances fraud transaction detection rates but also adapts to efficient processing demands in big data environments, providing new technical insights for financial risk control [11,12].

II. QUANTUM-SMOTE

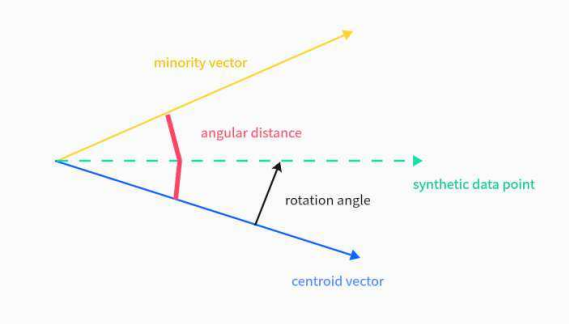

Figure 1. Quantum-SMOTE principle.

Quantum-SMOTE is an innovative approach that leverages quantum computing to enhance traditional Synthetic Minority Oversampling Techniques (SMOTE), specifically addressing class imbalance challenges in machine learning. Unlike conventional SMOTE methods that rely on K-nearest neighbor (KNN) and Euclidean distance to generate synthetic samples through neighborhood interpolation, Quantum-SMOTE harnesses the unique advantages of quantum computing. By employing quantum swap tests and quantum rotation operations, it efficiently processes high-dimensional data, reduces quantum resource requirements, and significantly improves classification performance.

Through amplitude encoding and compact exchange tests, Quantum-SMOTE compresses high-dimensional classical data into a logarithmic number of qubits. This efficient representation enables the algorithm to handle high-dimensional datasets, whereas traditional SMOTE may become inefficient in high-dimensional spaces due to "dimensional catastrophe".

Quantum-SMOTE employs low-depth quantum circuits (with fewer qubit operations), effectively reducing the impact of quantum noise and decoherence. This is particularly crucial for addressing the limitations of current quantum hardware (noise-integrated quantum devices, NISQ) to ensure practical algorithm implementation.

Traditional SMOTE relies on K-nearest neighbor selection and interpolation, which may introduce noise or generate unnatural samples. Quantum-SMOTE directly generates synthetic points from individual minority class samples through quantum rotation, reducing dependence on neighboring samples and improving the quality of generated samples.

Quantum-SMOTE introduces hyperparameters such as rotation angle, minority class ratio, and segmentation factor, allowing users to optimize synthetic data generation according to the characteristics of the dataset. These hyperparameters provide flexibility to adapt to different classification tasks.

Quantum-SMOTE employs K-means clustering to group minority samples into clusters before synthesizing samples. The clustering aims to identify natural data partitions, ensuring synthetic data points are generated within relevant subgroups to enhance their representativeness.

The K-means clustering algorithm assigns data points to clusters by minimizing intra-cluster variance, with each cluster represented by its centroid—the average value of all data points within that cluster. This approach ensures synthetic data points are generated based on local cluster characteristics, effectively avoiding the limitations of uniform sampling across minority class distributions.

III. QUANTUM SUPPORT VECTOR MACHINE

The Quantum Support Vector Machine (QSVM) is a quantum computing-based supervised learning algorithm designed for classification tasks. By integrating the theoretical framework of classical Support Vector Machines (SVM) with quantum computing's computational advantages, it enhances performance through quantum feature mapping and quantum kernel methods. This approach demonstrates particular effectiveness in handling high-dimensional data and nonlinear classification challenges.

QSVM combines the kernel method of classical SVM with quantum computing. It constructs kernel functions through quantum feature mapping and improves efficiency by utilizing the parallelism and high-dimensional state space processing capabilities of quantum computing. The core of QSVM lies in the generation of quantum kernel functions and the optimization of quantum circuits.

A. Quantum Feature Mapping

QSVM employs quantum feature mapping to encode classical data into quantum states. This process is implemented through quantum circuits and typically consists of the following steps.

1) Data Encoding

Given classical data $x \in \mathbb{R}^d$, it is mapped into a quantum state. Common methods include Amplitude Encoding and Angle Encoding.

a) Amplitude Encoding

A d dimensional vector $x = (x_0, x_1, \dots, x_{d-1})$ is normalized and encoded into the amplitudes of $\log_2 d$ qubits:

$$x \mapsto |x\rangle = \sum_{i=0}^{d-1} \frac{x_i}{\|x\|} |i\rangle \tag{1}$$

b) Angle Encoding

Each data feature is encoded as a rotation angle of a qubit, for instance using the R_y gate:

$$|\phi(x_i)\rangle = \bigotimes_{j=1}^{M} R_y\,(x_{i,j})|0\rangle \tag{2}$$

2) Quantum Circuit Construction

A parameterized quantum circuit $U_\phi(x)$ is designed to transform the initial state $|0\rangle^{\otimes n}$ into the feature state:

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$$|\phi(x)\rangle = U_\phi(x)|0\rangle^{\otimes n} \tag{3}$$

Such circuits usually combine single-qubit rotation gates with multi-qubit entangling gates (e.g., CNOT) to capture complex correlations among features.

3) High-Dimensional Embedding

Quantum feature mapping embeds classical data into a Hilbert space of dimension 2^n, where is the number of qubits. This provides a much richer representation than conventional kernel methods.

B. Quantum Kernel Function

The kernel function of QSVM is defined as the inner product between two quantum feature states:

$$K(x_i, x_j) = |\langle\phi(x_i)|\phi(x_j)\rangle|^2 \tag{4}$$

Prepare two quantum states $|\phi(x_i)\rangle$ and $|\phi(x_j)\rangle$, together with an ancillary control qubit $|0\rangle$.

Apply a Hadamard gate to the control qubit:

$$H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \tag{5}$$

Apply a controlled-SWAP (Fredkin) gate: when the control qubit is $|1\rangle$, the two states are swapped.

Apply another Hadamard gate to the control qubit and measure. The probability of obtaining $|0\rangle$ is:

$$P_0 = \frac{1}{2}\left(1 + |\langle\phi(x_i)|\phi(x_j)\rangle|^2\right) \tag{6}$$

Thus, the kernel value is computed as:

$$K(x_i, x_j) = 2P(0) - 1 \tag{7}$$

C. QSVM Training and Classification

1) Quantum Kernel Matrix Construction

The quantum computer evaluates the kernel entries for all pairs of training samples:

$$K_{ij} = K(x_i, x_j) = |\langle\phi(x_i)|\phi(x_j)\rangle|^2 \tag{8}$$

2) Classical Optimization

The kernel matrix is passed to a classical SVM solver, which solves the dual optimization problem. This yields the support vectors defined by the coefficients α_i .The trained decision function is expressed as:

$$f(x) = \text{sign}\left(\sum_{i=1}^{m} \alpha_i\, y_i K(x_i, x) + b\right) \tag{9}$$

3) Prediction

For a new data point x, the quantum circuit encodes it into $|\phi(x)\rangle$, evaluates the kernel with the support vectors, and the classifier outputs the prediction.

IV. EXPERIMENTAL SIMULATION

A. Data preparation

# Time Number of seconds elapsed between this transaction and the first transaction in the dataset	# V1 may be result of a PCA Dimensionality reduction to protect user identities and sensitive features(v1-v28)	# V2	# V3	# V4
0 173k	-56.4 2.45	-72.7 22.1	-48.3 9.38	-5.68 16.9
0	-1.3598071336738	-0.0727811733098497	2.53634673796914	1.37815522427443
0	1.19185711131486	0.26615871205963	0.166480113335321	0.448154078460911
1	-1.35835406159823	-1.34016307473609	1.77320934263119	0.379779593034328
1	-0.966271711572087	-0.185226008082898	1.79299333957872	-0.863291275036453
2	-1.15823309349523	0.877736754848451	1.548717846511	0.403833933955121
2	-0.425965884412454	0.960523844882985	1.14110934232219	-0.168252079768382
4	1.22965763450793	0.141003507049326	0.0453707735899449	1.20261273673594
7	-0.644269442348146	1.41796354547385	1.0743803763556	-0.492199018495015
7	-0.894286008220282	0.286157196276544	-0.113192212729871	-0.271526130088604
9	-0.338261755242575	1.11959337641566	1.04436655157316	-0.222187276738296
10	1.44904378114715	-1.17633882535966	0.913859832832795	-1.37566665499943
10	0.384978215188095	0.616109459176472	-0.874299782595052	-0.0940186259679115
10	1.249998742053	-1.22163680921816	0.383930151282291	-1.23489868766892

Figure 2. Data structure.

In this study, we utilized the credit card fraud detection dataset from Kaggle, which captures transaction patterns over two days. Among 284,807 transactions, 492 were identified as fraudulent. The dataset exhibits severe imbalance, with fraudulent transactions constituting merely 0.172% of all transactions. It contains only numerical input variables derived from Principal Component Analysis (PCA). Features V1, V2,..., V28 represent PCA-derived principal components, while the sole non-PCA-transformed features are "time" and "amount". The "time" feature measures seconds elapsed between each transaction and the dataset's first entry. The "amount" feature denotes transaction values, which can be utilized for example-based cost-sensitive learning. The "Class" feature serves as the response variable, taking value 1 in fraudulent cases and 0 otherwise.

In such an extremely unbalanced data set, the classifier is usually overwhelmed by the most class, that is, the data of non-fraudulent transactions, resulting in the failure of the classifier to predict the fraudulent transactions well. In this work, the minority class (fraud) samples are generated by Quantum-SMOTE to increase its proportion.

The dataset contains 30 features (V1-V28, Time, Amount), while Qiskit's QSVM employs a quantum bit count that correlates with feature dimensions. High-dimensional features increase quantum circuit complexity and computational costs, making feature selection crucial. This study utilizes KBest to select features based on their scores, compares recall/accuracy ratios, and selects the top 10 features above V19 as training samples according to the

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ranking.

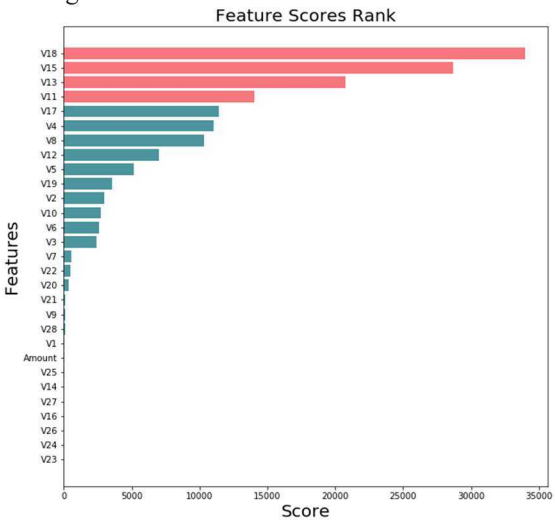

Figure 3. Feature scores rank.

B. *Experimental results*

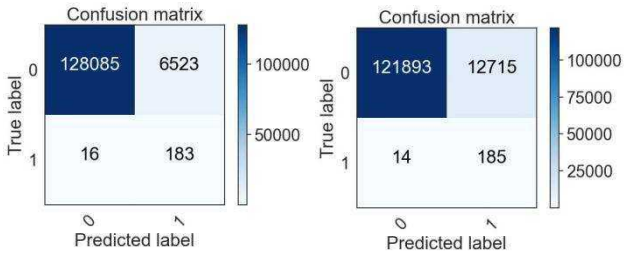

Figure 4. The confusion matrix of QSVM and SVM.

This study employed two distinct data balancing techniques for preprocessing the original dataset, subsequently training a 10-qubit quantum support vector machine (QSVM) classification model. We first utilized Quantum-SMOTE quantum synthetic minority over-sampling to generate synthetic fraud samples, thereby creating a balanced training dataset. For comparative analysis, we also applied traditional undersampling methods: selecting the first 15,000 samples from the original dataset as the basic training set containing only 293 fraudulent instances (positive examples), while randomly assigning 300 non-fraudulent samples (negative examples) to form a control training set. During evaluation, we trained QSVM models using both Quantum-SMOTE-processed and undersampled datasets, systematically comparing their performance across key metrics including accuracy, recall, precision, and F1 score. Table 1 provides detailed comparative results of QSVM classification performance under these two data preprocessing approaches.

In the era of Noisy Intermediate Scale Quantum (NISQ) computing, the number of qubits is severely limited, which directly impacts the expressive power of quantum machine learning models. To investigate how different qubit scales affect classification performance, this study trains QSVM models using 4, 8, and 10 qubits respectively. Within the Qiskit framework, we employ ZZFeatureMap for feature mapping, where the number of qubits equals the feature dimension of input data. Consequently, we adjusted the dataset dimensions to align with 4, 8, and 10 features respectively.

To fairly evaluate the performance of quantum models compared to classical methods, we trained classical Support Vector Machines (SVMs)using data with three feature dimensions (4, 8, and 10), and assessed their classification capabilities on identical test sets. Table 2 presents a performance comparison between quantum SVM (QSVM) and classical SVM across different feature dimensions, including key metrics such as accuracy, recall rate, precision, and F1 score. This analysis reveals the advantages and limitations of quantum computing under finite-bit constraints.

TABLE I. EVALUATION INDICATORS OF MODEL TRAINING AFTER QUANTUM-SMOTE AND UNDERSAMPLING PROCESSING

	qubits	accuracy	F1-score	recall	AUC
Quantum-SMOTE	10	98.8	0.962	0.945	0.992
Under-sampling	10	92.4	0.907	0.881	0.972

TABLE II. EVALUATION INDICATORS OF QSVM AND SVM TRAINING UNDER DIFFERENT FEATURE DIMENSIONS

	qubits	accuracy	F1-score	recall	AUC
QSVM	4	92.2	0.902	0.894	0.971
	8	96.7	0.956	0.941	0.983
	10	98.8	0.962	0.945	0.992
SVM	4	91.4	0.897	0.871	0.969
	8	94.3	0.932	0.921	0.973
	10	95.1	0.933	0.923	0.979

In-depth analysis of experimental data demonstrates that quantum machine learning technology exhibits significant advantages in credit card fraud detection tasks. When comparing Quantum SMOTE with traditional undersampling methods using 10 qubits, Quantum SMOTE achieved an accuracy rate of 98.8%, outperforming the 92.4% accuracy rate of traditional undersampling methods by 6.4 percentage points. This performance gap is evident not only in accuracy but also in other key metrics such as F1-score (0.962 vs. 0.907), recall (0.945 vs. 0.881), and AUC (0.992 vs. 0.972). These comprehensive improvements indicate that Quantum SMOTE better preserves the true distribution characteristics of minority class samples during generation, effectively avoiding the information loss issues associated with traditional undersampling methods.

A comparative analysis of QSVM and traditional SVMs across different qubit configurations reveals an intriguing trend: While both methods show performance improvements with increasing qubit numbers, QSVM demonstrates more pronounced enhancements. At 4 qubits, QSVM achieves 92.2% accuracy—just 0.8 percentage points higher than SVM's 91.4%. The gap widens to 2.4 percentage points (96.7% vs. 94.3%) when scaling up to 8 qubits, and expands further to 3.7 percentage points (98.8% vs. 95.1%) at 10 qubits. This performance disparity that grows with quantum resource expansion vividly illustrates the scalability advantage of quantum computing in handling complex pattern recognition tasks.

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Notably, the 0.992 AUC score achieved with the top-tier configuration (Quantum-SMOTE + 10-bit QSVM) approaches the performance of a perfect classifier, demonstrating exceptional capability in distinguishing fraudulent from legitimate transactions. The model's F1-score of 0.962 and recall rate of 0.945 further highlight its ability to maintain high precision while accurately identifying minority classes (fraudulent transactions). This is particularly crucial for real-world financial applications, as the cost of missing fraudulent transactions often far exceeds that of misidentifying legitimate ones.

V. CONCLUSION

This work proposes a credit card fraud detection model based on quantum support vector machine (QSVM) and quantum synthetic minority over sampling technology (Quantum-SMOTE), which is verified by experiments to have significant advantages in dealing with unbalanced data sets and high-dimensional feature classification tasks.

Compared with traditional under-sampling methods, Quantum-SMOTE significantly improves data imbalance by generating high-quality synthetic minority class samples through quantum swapping tests and quantum rotation. Experimental results demonstrate that models trained with Quantum-SMOTE outperform conventional approaches in key metrics including accuracy (98.8%), F1-score (0.962), recall (0.945), and AUC (0.992). This validates its potential in maintaining data distribution authenticity while enhancing classification performance.

QSVM enhances feature separability in high-dimensional Hilbert space through quantum feature mapping and quantum kernel computation. As the number of qubits increases from 4 to 10, QSVM demonstrates significantly better performance than classical SVM, with the accuracy gap reaching 3.7 percentage points in the 10-qubit configuration. This highlights the scalability advantage of quantum computing in handling complex pattern recognition tasks.

Under hardware constraints in the NISQ era, this study achieved near-perfect classifier performance (AUC=0.992) through optimized qubits and feature selection. This breakthrough provides innovative technical solutions for financial risk control, particularly demonstrating significant practical value in high-precision credit card fraud detection scenarios with low false-negative rates.

While this study has yielded significant results, the application of quantum machine learning in financial domains remains in its exploratory phase. Future work should focus on optimizing quantum circuit design, developing more efficient quantum feature mapping techniques, and deploying models on real quantum hardware to validate their practicality and robustness across large-scale datasets.

This study pioneers the application of quantum machine learning in credit card fraud detection, offering a novel technical framework for this field while laying groundwork for quantum computing's potential applications in fintech. With continuous advancements in quantum hardware, quantum machine learning is poised to demonstrate its unique advantages across increasingly complex tasks.

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